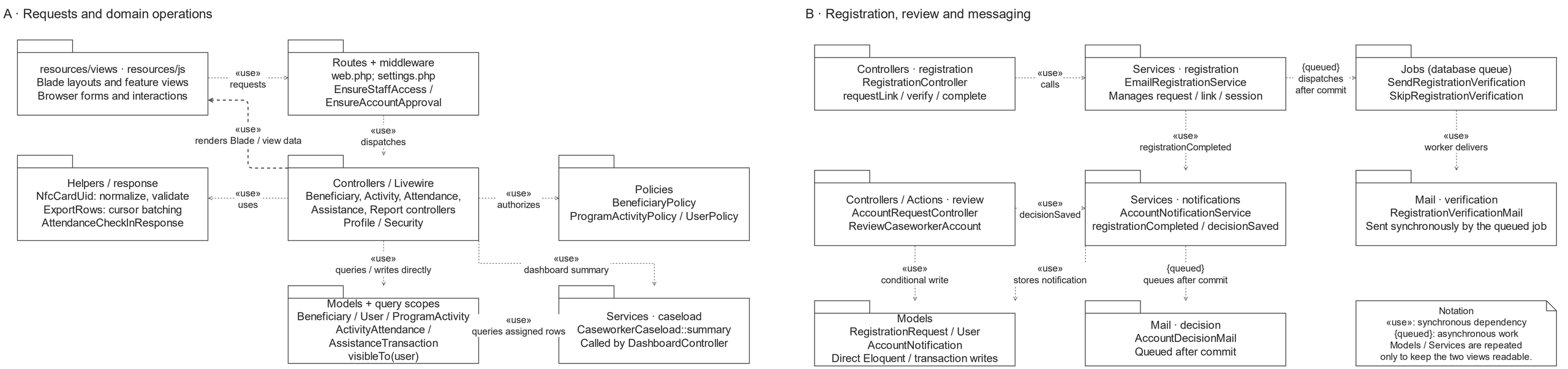
Package Diagram of the Proposed System

Packages organize the implementation. Dashed open arrows are UML dependencies; labels distinguish synchronous use from queued dispatch after commit.

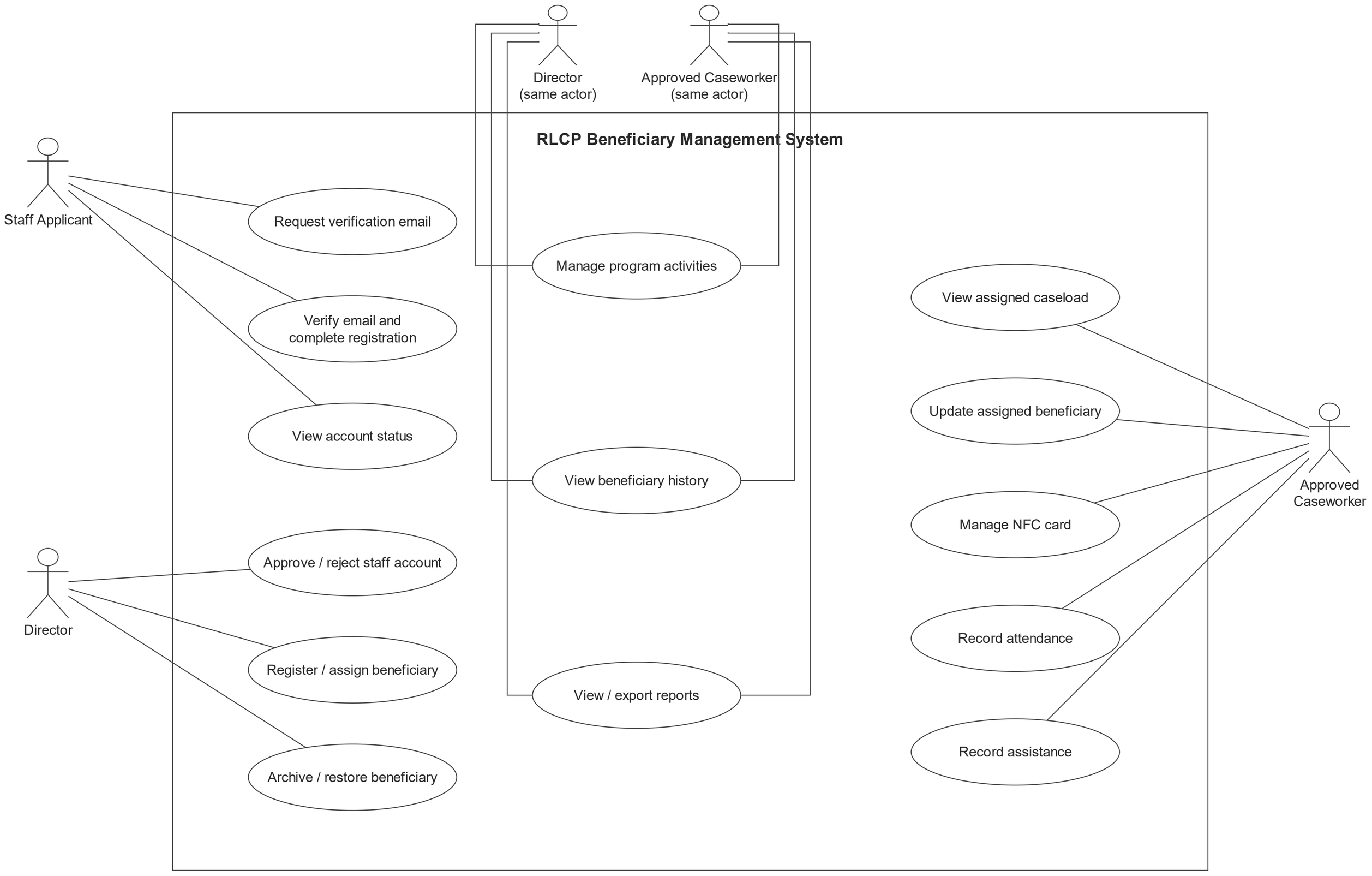


Source: inventory.md AUTH / REG / ACC rules; routes, controllers, Livewire settings, services, jobs and mail. The two panels expand the same application.

Figure 1. Package Diagram of the Proposed System

Use Case Diagram of the Proposed System

Use cases describe user goals. Staff roles require verified email; Caseworkers also require approval. Beneficiaries are records, not users of the software.

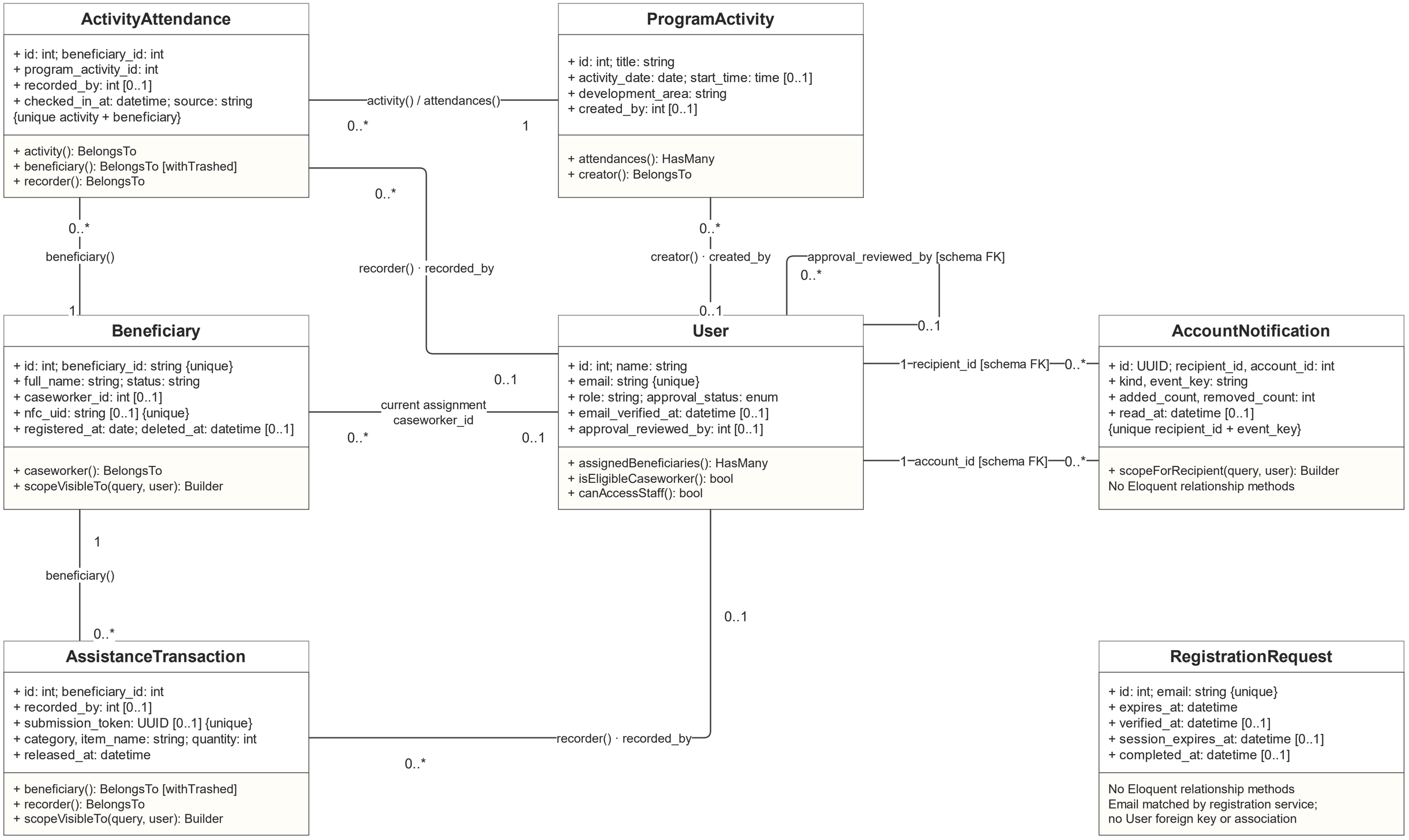


Director may also update beneficiaries, manage NFC cards, record attendance and assistance across the organization.
Caseworker beneficiary operations and reports are restricted to current assignments; attendance and assistance require active records.
Shared actor symbols denote the same roles, not additional accounts. Source: User, BeneficiaryPolicy, ProgramActivityPolicy and inventory.md.

Figure 2. Use Case Diagram of the Proposed System

Class Diagram of the Proposed System

Current caseworker assignment governs access. Historical creator / recorder fields preserve who performed an action. Multiplicities appear at both ends of every association.

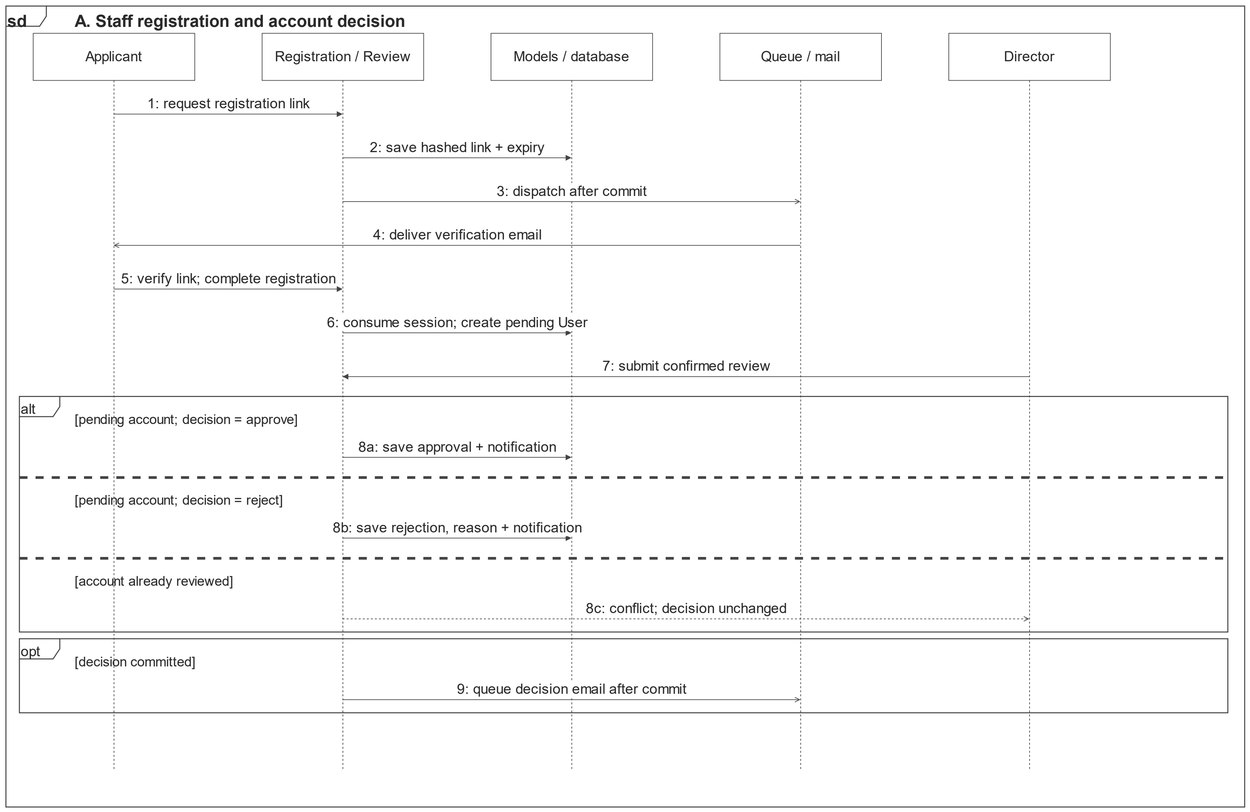


+ denotes public Eloquent attributes / operations; [0..1] denotes an optional value. Only relevant attributes and methods are shown.
Association labels name implemented methods or schema foreign keys. Assignment controls access; recorder / creator links preserve historical authorship.
RegistrationRequest is matched by email in application logic; no User association exists. Source: app/Models and migration declarations.

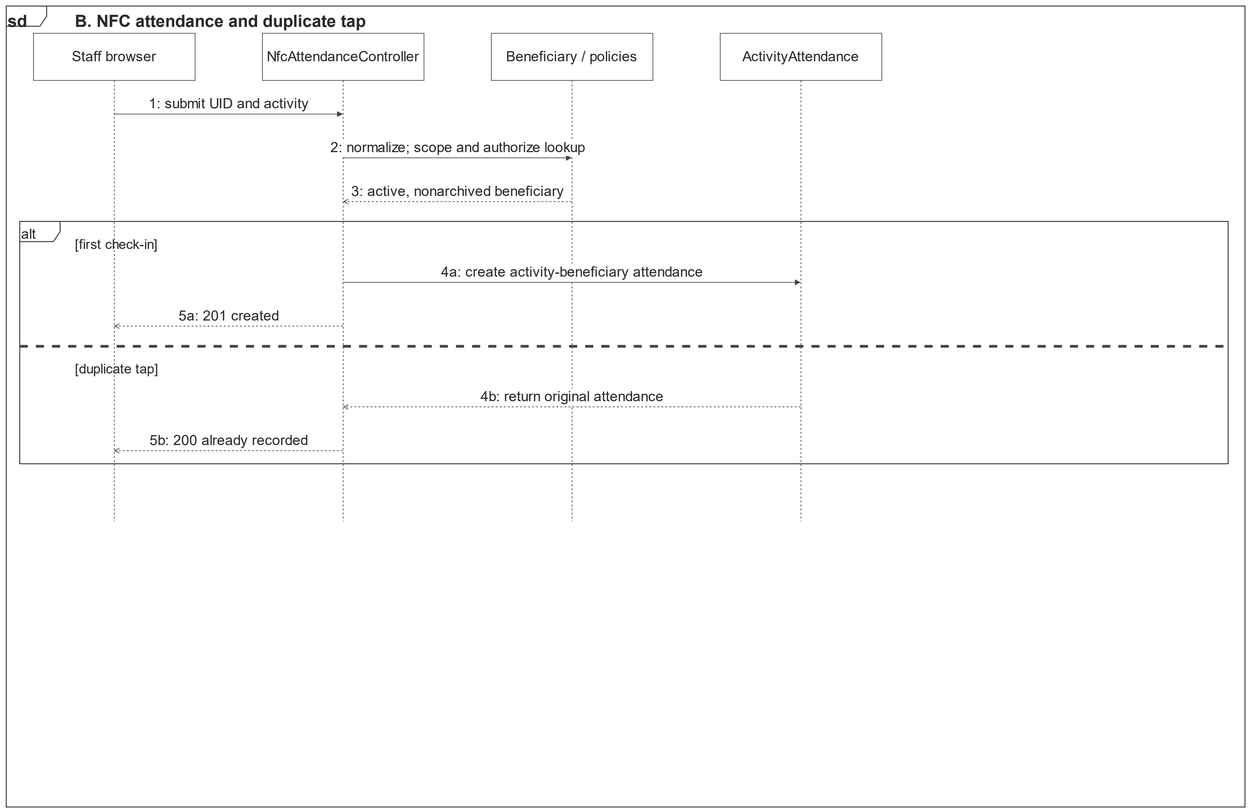
Figure 3. Class Diagram of the Proposed System

Sequence Diagram of the Proposed System

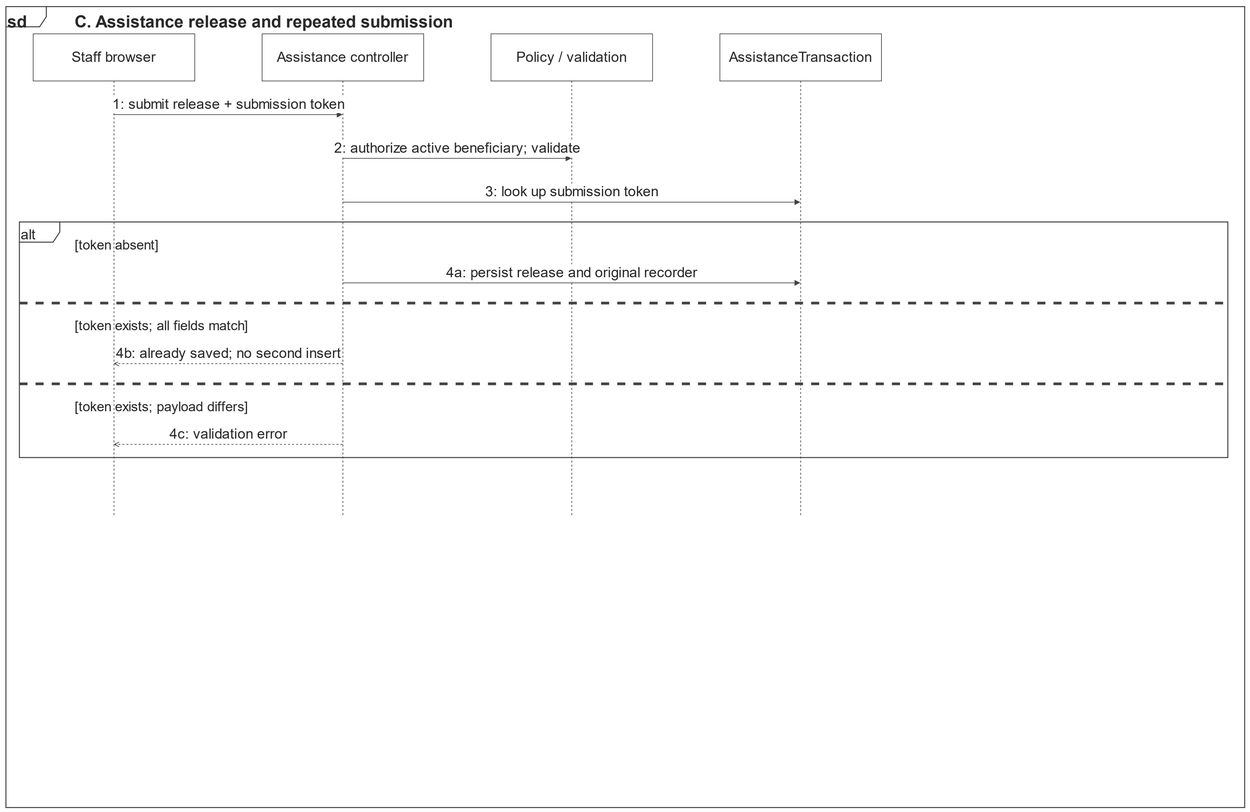
Each interaction is independent. Time flows down; solid filled arrows are calls, solid open arrows are asynchronous messages, and dashed open arrows are replies.



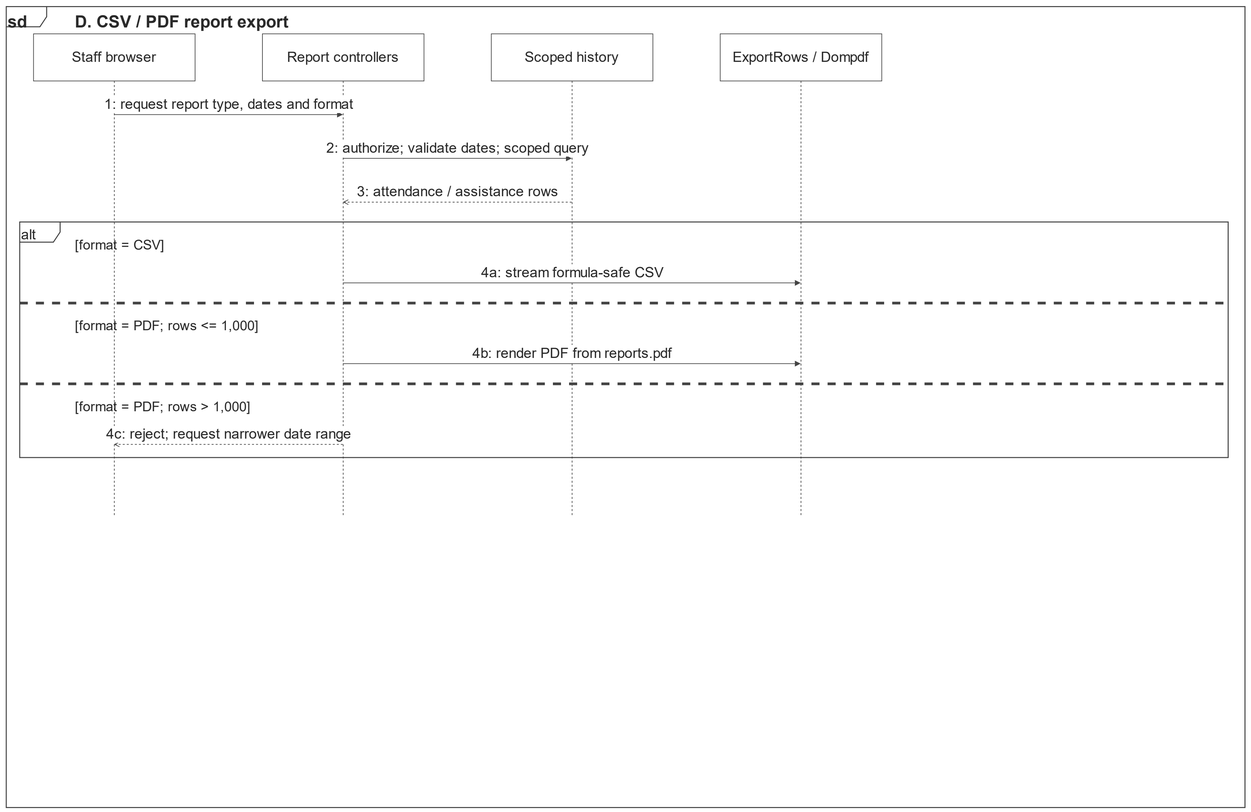
Invalid / expired / used links or sessions fail. New users are email-verified but pending. Approval enables staff access; rejection retains restrictions.



Invalid, inaccessible or inactive records fail before persistence. firstOrCreate and a composite unique key prevent duplicates. Endpoint source = nfc_test.



Replay comparison includes recorder and UTC release time. A token-uniqueness race rereads the original row. Attendance is not a prerequisite.

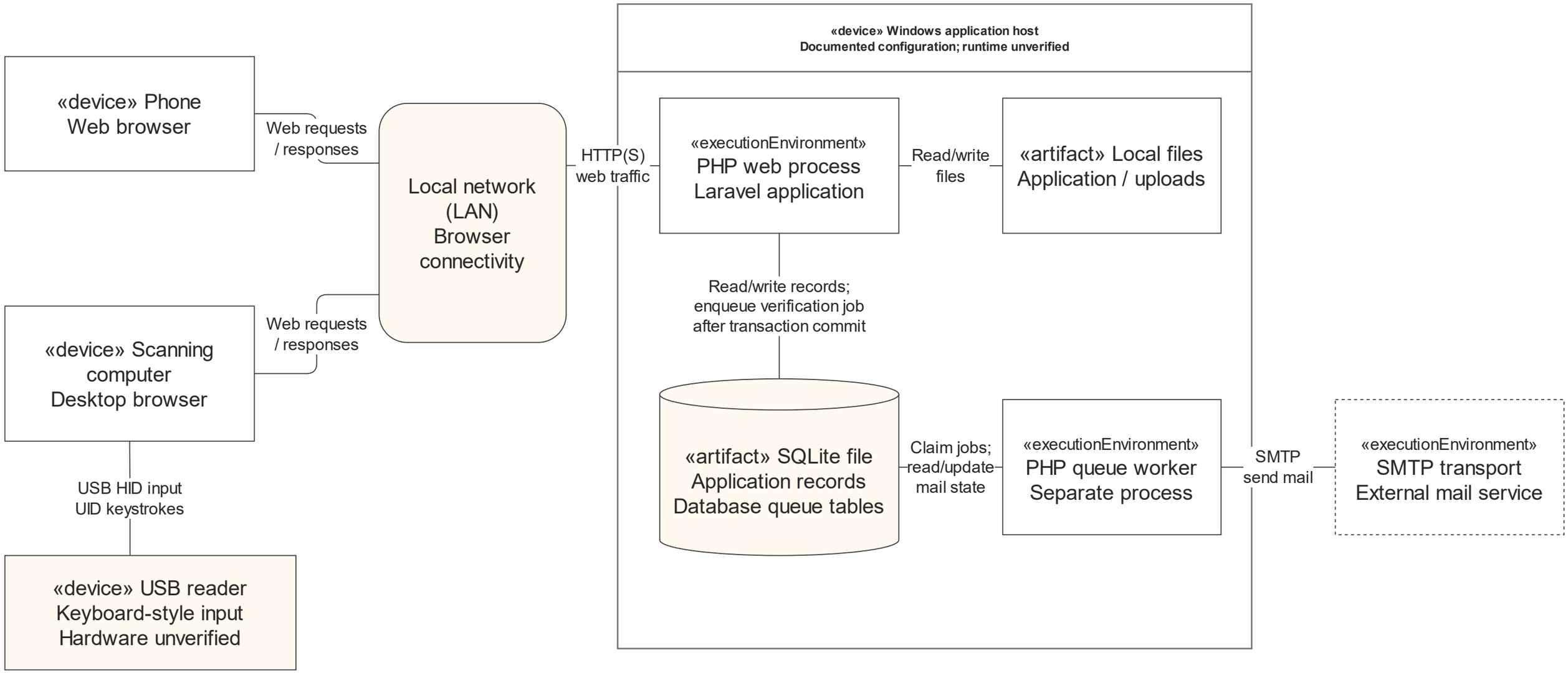


Successful exports download with private, no-store headers. Queries are read-only. Activity-specific CSV uses the selected activity and optional search.

Figure 4. Sequence Diagram of the Proposed System

Deployment Diagram of the Proposed System

Documented Windows LAN topology. This diagram describes the setup; current devices, processes, configuration and service availability were not inspected.



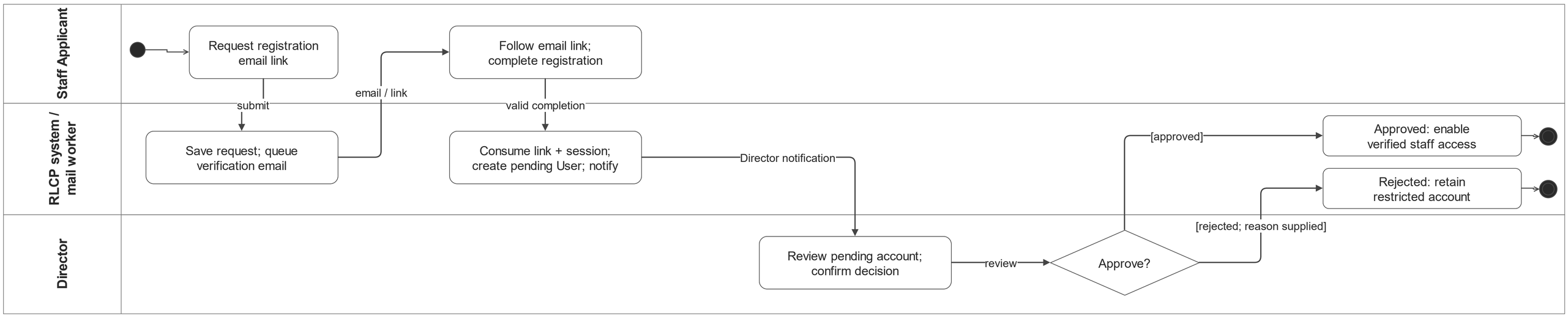
Evidence: docs/local-lan-mail-queue.md, documented LAN setup and inventory.md. Separate web and queue PHP processes share local persistence.
Database-queued verification and after-commit decision mail use SMTP. USB keyboard input is an assumption; hardware was not tested. No LAN addresses shown.

Figure 5. Deployment Diagram of the Proposed System

Process Flow of the Proposed System

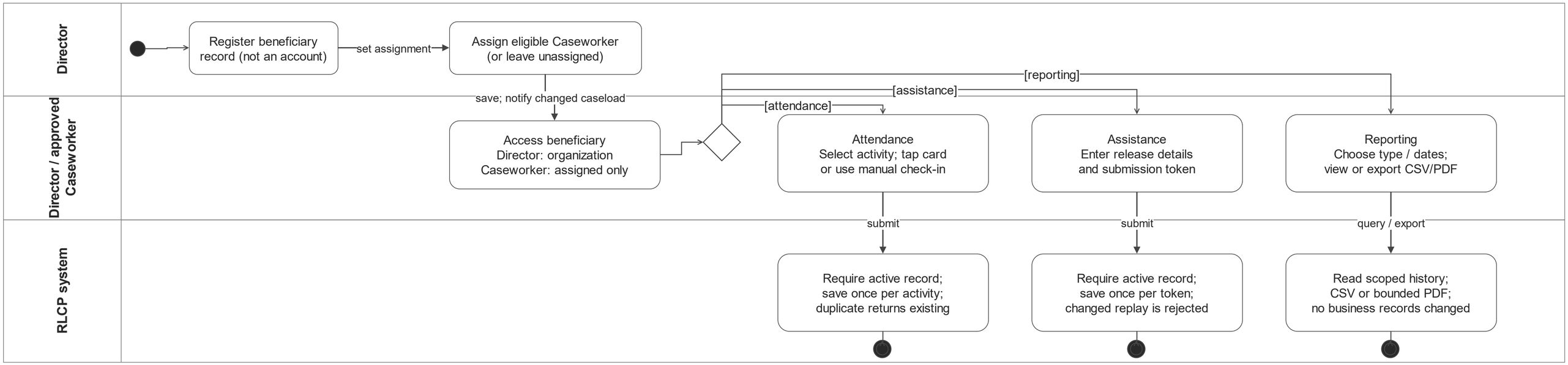
Lanes identify responsibility. Staff onboarding is separate from beneficiary records. Attendance, assistance and reporting are independent choices; none requires another.

A · Staff onboarding and approval



Invalid / expired links or sessions fail. Both decisions store notifications and queue mail after commit. A repeated review returns conflict.

B · Beneficiary registration, assignment and program operations



Choose one operation per interaction; the user can return later for any other operation. No attendance-before-assistance requirement exists.
Activity definitions are organization-wide; beneficiary records follow role / assignment scope. Director alone registers, assigns, archives and restores.
Black circles denote starts; bullseyes denote completed interactions; bracketed labels are decision guards. Source: inventory.md and current controllers.

Figure 6. Process Flow of the Proposed System